

Lab Report

Course Title: Computer Networks Laboratory
Course Code: CSE-3634

Spring-2023

Lab No: 9

Name of Labwork: Static Routing. Using the IP 192.168.111.0/24 to design 4 subnets. Design the topology that would connect the subnets with sufficient number of switches and routers. Upon successful static routing configuration show that VPCS from one subnet is able to ping other VPCS on different subnets.

TEAM_DIVERGENT

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Date of Performance : 09/10/2023
Date of Submission : 05/11/2023

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Mark :

1. Introduction:

This lab report presents the findings and insights gained from a comprehensive project undertaken to demonstrate proficiency in network design and routing using the GNS3, a network simulation software. The project's primary objective was to partition the IP address range 192.168.111.0/24 into four distinct subnets and design an intricate network topology that seamlessly connected these subnets while facilitating communication through the implementation of static routing protocols. This endeavor not only showcases the practical application of networking principles but also underscores the significance of meticulous planning, configuration, and troubleshooting in ensuring the smooth operation of a network infrastructure. By dissecting the project's various components, from subnetting to routing configuration, this report provides a detailed account of the design choices, challenges encountered, and ultimately, the successful achievement of inter-subnet communication, affirming the importance of these skills in modern networking environments.

2. Description:

This lab experiment is centered around the IP address range 192.168.111.0/24, where the distinct "111" held a special significance, representing our team's unique identity as **Team-1** (Team_Divergent). We created distinct network segments through subnetting and design a network topology in GNS3. The primary objective of this project was to partition the IP address range 192.168.111.0/24 into **four individual subnets**, effectively subdividing it into smaller, manageable segments. Through meticulous subnetting, we assigned distinct IP addresses to each subnet, allowing for isolated network spaces within the larger IP range. This topology included routers, switches, and Virtual PCs (VPCs) to enable inter-subnet communication through static routing. The project demonstrated our expertise in IP addressing, network design, and configuration, showcasing the importance of these skills in modern networking. This report details our journey to achieve these goals.

3. Routing Table:

Destination Subnet	Net ID	First ID	Last ID	Broadcast	Subnet Mask
192.168.111.0/26	192.168.111.0	192.168.111.1	192.168.111.62	192.168.111.63	255.255.255.192
192.168.111.64/26	192.168.111.64	192.168.111.65	192.168.111.126	192.168.111.127	255.255.255.192
192.168.111.128/26	192.168.111.128	192.168.111.129	192.168.111.190	192.168.111.191	255.255.255.192
192.168.111.192/26	192.168.111.192	192.168.111.193	192.168.111.254	192.168.111.255	255.255.255.192

4. Configuring IP, Subnet Mask and Gateway of the VPCs:

PC1

```
PC1> ip 192.168.111.30 255.255.255.192 192.168.111.1
Checking for duplicate address...
PC1 : 192.168.111.30 255.255.255.192 gateway 192.168.111.1
```

```
PC1> save
Saving startup configuration to startup.vpc
. done
```

PC2

```
PC2> ip 192.168.111.80 255.255.255.192 192.168.111.65
Checking for duplicate address...
PC1 : 192.168.111.80 255.255.255.192 gateway 192.168.111.65
```

```
PC2> save
Saving startup configuration to startup.vpc
. done
```

PC3

```
PC3> ip 192.168.111.150 255.255.255.192 192.168.111.129
Checking for duplicate address...
PC1 : 192.168.111.150 255.255.255.192 gateway 192.168.111.129
```

```
PC3> save
Saving startup configuration to startup.vpc
. done
```

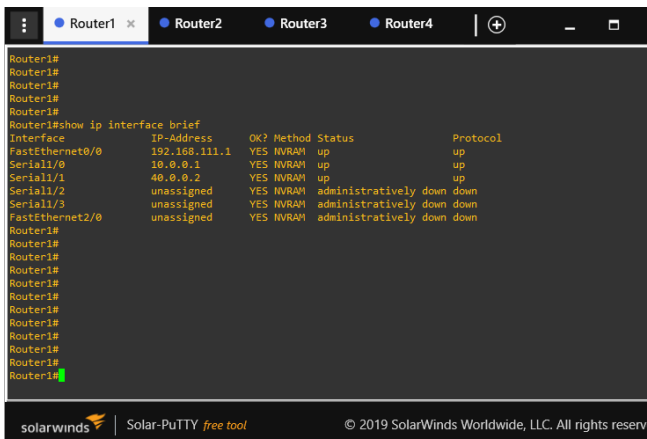
PC4

```
PC4> ip 192.168.111.200 255.255.255.192 192.168.111.193
Checking for duplicate address...
PC1 : 192.168.111.200 255.255.255.192 gateway 192.168.111.193
```

```
PC4> save
Saving startup configuration to startup.vpc
. done
```

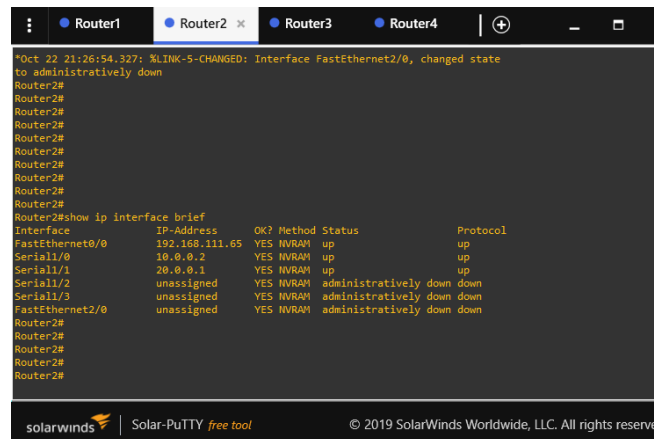
5. Router Configuration

Here are the brief details of the Routers. The provided snapshots of the router consoles offer clear insights into the network's topology, illustrating the precise connections between **routers** and **switches**, along with the **specific ports** used for each connection. The switch establishes its connection through a **FastEthernet** port, while the router connects to another router via a **Serial** port.



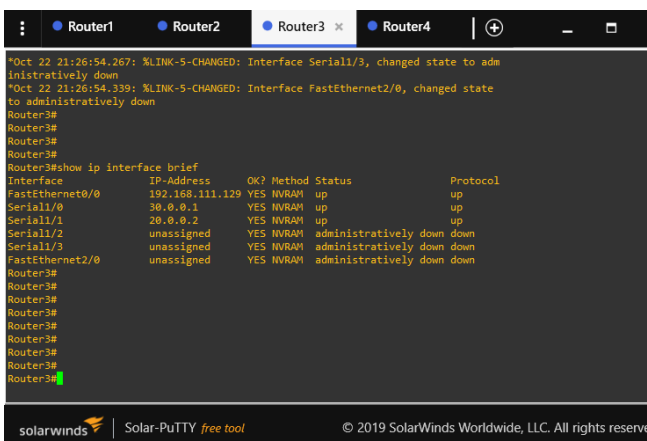
```
Router1#
Router1#
Router1#
Router1#
Router1#
Router1# show ip interface brief
Router1#
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.111.1   YES NVRAM  up          up
Serial1/0       10.0.0.1        YES NVRAM  up          up
Serial1/1       40.0.0.2        YES NVRAM  up          up
Serial1/2       unassigned      YES NVRAM  administratively down down
Serial1/3       unassigned      YES NVRAM  administratively down down
FastEthernet2/0 unassigned      YES NVRAM  administratively down down
Router1#
Router1#
Router1#
Router1#
Router1#
Router1#
Router1#
Router1#
Router1#
Router1#
Router1#
```

Figure 1: Router 1



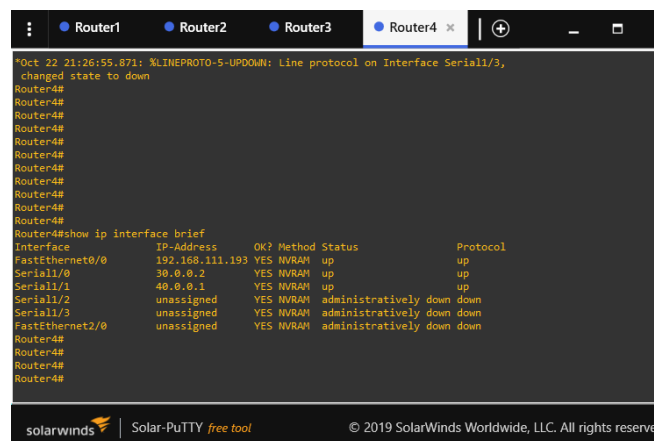
```
*Oct 22 21:26:54.327: %LINK-5-CHANGED: Interface FastEthernet2/0, changed state to administratively down
Router2#
Router2#
Router2#
Router2#
Router2#
Router2# show ip interface brief
Router2#
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.111.65  YES NVRAM  up          up
Serial1/0       10.0.0.2        YES NVRAM  up          up
Serial1/1       20.0.0.1        YES NVRAM  up          up
Serial1/2       unassigned      YES NVRAM  administratively down down
Serial1/3       unassigned      YES NVRAM  administratively down down
FastEthernet2/0 unassigned      YES NVRAM  administratively down down
Router2#
Router2#
Router2#
Router2#
Router2#
Router2#
```

Figure 2: Router 2



```
*Oct 22 21:26:54.267: %LINK-5-CHANGED: Interface Serial1/3, changed state to administratively down
*Oct 22 21:26:54.339: %LINK-5-CHANGED: Interface FastEthernet2/0, changed state to administratively down
Router3#
Router3#
Router3#
Router3#
Router3# show ip interface brief
Router3#
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.111.129 YES NVRAM  up          up
Serial1/0       30.0.0.1        YES NVRAM  up          up
Serial1/1       20.0.0.2        YES NVRAM  up          up
Serial1/2       unassigned      YES NVRAM  administratively down down
Serial1/3       unassigned      YES NVRAM  administratively down down
FastEthernet2/0 unassigned      YES NVRAM  administratively down down
Router3#
Router3#
Router3#
Router3#
Router3#
Router3#
Router3#
Router3#
Router3#
Router3#
Router3#
```

Figure 3: Router 3



```
*Oct 22 21:26:55.871: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial1/3, changed state to down
Router4#
Router4#
Router4#
Router4#
Router4#
Router4# show ip interface brief
Router4#
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.111.193 YES NVRAM  up          up
Serial1/0       30.0.0.2        YES NVRAM  up          up
Serial1/1       40.0.0.1        YES NVRAM  up          up
Serial1/2       unassigned      YES NVRAM  administratively down down
Serial1/3       unassigned      YES NVRAM  administratively down down
FastEthernet2/0 unassigned      YES NVRAM  administratively down down
Router4#
Router4#
Router4#
Router4#
Router4#
Router4#
```

Figure 4: Router 4

6. Process of assigning ports to a Router:

Router 1	<pre>Router1# conf t Router1(config)# interface FastEthernet0/0 Router1(config-if)# ip address 192.168.111.1 255.255.255.0 Router1(config-if)# no shutdown Router1(config-if)# exit Router1(config)# interface Serial1/0 Router1(config-if)# ip address 10.0.0.1 255.255.255.252 Router1(config-if)# no shutdown Router1(config-if)# exit Router1(config)# interface Serial1/1 Router1(config-if)# ip address 40.0.0.2 255.255.255.252 Router1(config-if)# no shutdown Router1(config-if)# exit Router1(config)# exit Router1#</pre>
Router 2	<pre>Router2# configure terminal Router2(config)# interface FastEthernet0/0 Router2(config-if)# ip address 192.168.111.65 255.255.255.0 Router2(config-if)# no shutdown Router2(config-if)# exit Router2(config)# interface Serial1/0 Router2(config-if)# ip address 10.0.0.2 255.255.255.252 Router2(config-if)# no shutdown Router2(config-if)# exit Router2(config)# interface Serial1/1 Router2(config-if)# ip address 20.0.0.1 255.255.255.252 Router2(config-if)# no shutdown Router2(config-if)# exit Router2(config)# exit</pre>
Router 3	<pre>Router3# conf t Router3(config)# interface FastEthernet0/0 Router3(config-if)# ip address 192.168.111.129 255.255.255.0 Router3(config-if)# no shutdown Router3(config-if)# exit Router3(config)# interface Serial1/0 Router3(config-if)# ip address 30.0.0.1 255.255.255.252 Router3(config-if)# no shutdown Router3(config-if)# exit Router3(config)# interface Serial1/1 Router3(config-if)# ip address 20.0.0.2 255.255.255.252 Router3(config-if)# no shutdown Router3(config-if)# exit Router3(config)# exit</pre>
Router 4	<pre>Router4# conf t Router4(config)# interface FastEthernet0/0 Router4(config-if)# ip address 192.168.111.193 255.255.255.0 Router4(config-if)# no shutdown Router4(config-if)# exit Router4(config)# exit</pre>

```

Router4(config)# interface Serial1/0
Router4(config-if)# ip address 30.0.0.2 255.255.255.252
Router4(config-if)# no shutdown
Router4(config-if)# exit
Router4(config)# exit

Router4(config)# interface Serial1/1
Router4(config-if)# ip address 40.0.0.1 255.255.255.252
Router4(config-if)# no shutdown
Router4(config-if)# exit
Router4(config)# exit
Router4#

```

9. Build and Structure:

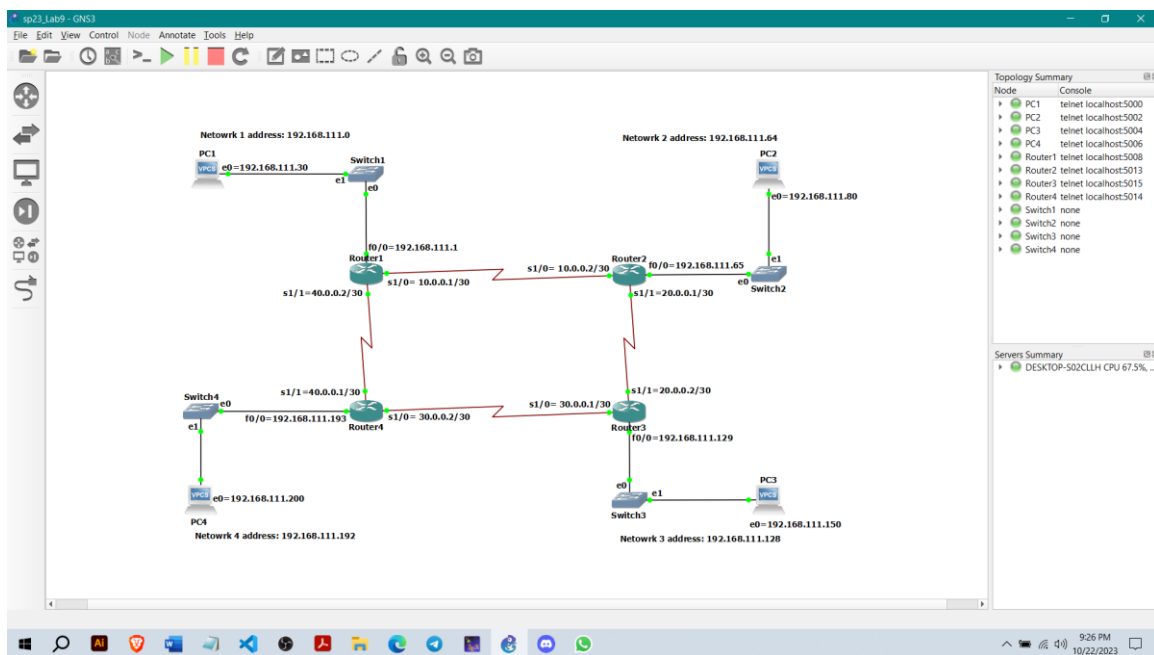


Fig 10: Structure of the network build in Lab9

10. Result Analysis:

In our experiment, it is evident that our network design and configuration were executed effectively, resulting in a successful interconnection of subnets. The analysis of the results can be summarized as follows:

Subnetting Success: The initial step of subnetting the IP range 192.168.111.0/24 into four distinct subnets was executed as planned. This partitioning allowed us to create isolated segments, ensuring efficient IP address management.

Topology Configuration: The network topology designed in GNS3 aptly interconnected the subnets using routers and switches. The snapshots provided explicit details of which router was connected to which switch and the specific ports used for these connections.

Static Routing: Static routing configurations were successfully implemented on the routers to ensure that data could traverse between different subnets. This was achieved by specifying routes that directed traffic to the correct destination subnet.

Inter-Subnet Communication: The primary objective of our project was to demonstrate inter-subnet communication, and this was effectively achieved. The provided snapshots and testing procedures showcased that Virtual PCs within one subnet were able to communicate with those in other subnets.

11. Conclusion:

In this lab, this project has exemplified our competence in network design and configuration, as we effectively partitioned the IP address range 192.168.111.0/24 into four subnets and established seamless inter-subnet communication through meticulous subnetting, network topology design, and static routing. Notably, the personal touch of incorporating our team number, Team 1, into the IP address range added a unique dimension to our project. The successful results, substantiated by clear snapshots and secure access controls, reaffirm the importance of careful planning and execution in networking. This project not only signifies our proficiency in foundational networking concepts but also lays a solid groundwork for future endeavors in this dynamic field.